Experimet 3

Study of network Cables and Crimping with RJ45 Connector

Aim

To Study of different types of Network cables and crimping with RJ45 Connector

Content

Cable type	Category	Maximum Data transmition	Application
UTP	Category 3	1a bps	10 Base T
	Category 5	upto 100 Mbps	Fast, Gigabit
	Category 5e	1e Gbps	Fast, Gigabit
STP	Category 6, 6a	10 Gbps	Gigabit, 10G widely used in data centres
SSTP	Category 7	10Gbps	Gigabit, 10G (100 m)
Coaxial cable	RG-6, RG-59 RG-11	10-100 Mbps	Speed of Signal is 500m, Television network
fibre optics cable	Single mode multi mode	100 Gbps	Maximum distance of fibre optic Cable

RJ45 is the standard connector for Ethernet cables. Two wiring standards exist: -TIA/EIA 568A: Green-white, Green, Orange-white, Brown -TIA/EIA 568B: orange-white, orange, Green-white, Blue, Blue-white, Green, Brown-white, Brown

Straight-through cable uses the same standard (568B-568B) on both ends and connects PC to switch. Cross over cable uses 586A on one end and 568B on the other-Connects PC to PC directly

Result

Both Straight-through and Cross-over Ethernet cables were successfully crimped and verified using a cable tester, All 8 pins showed proper continuity